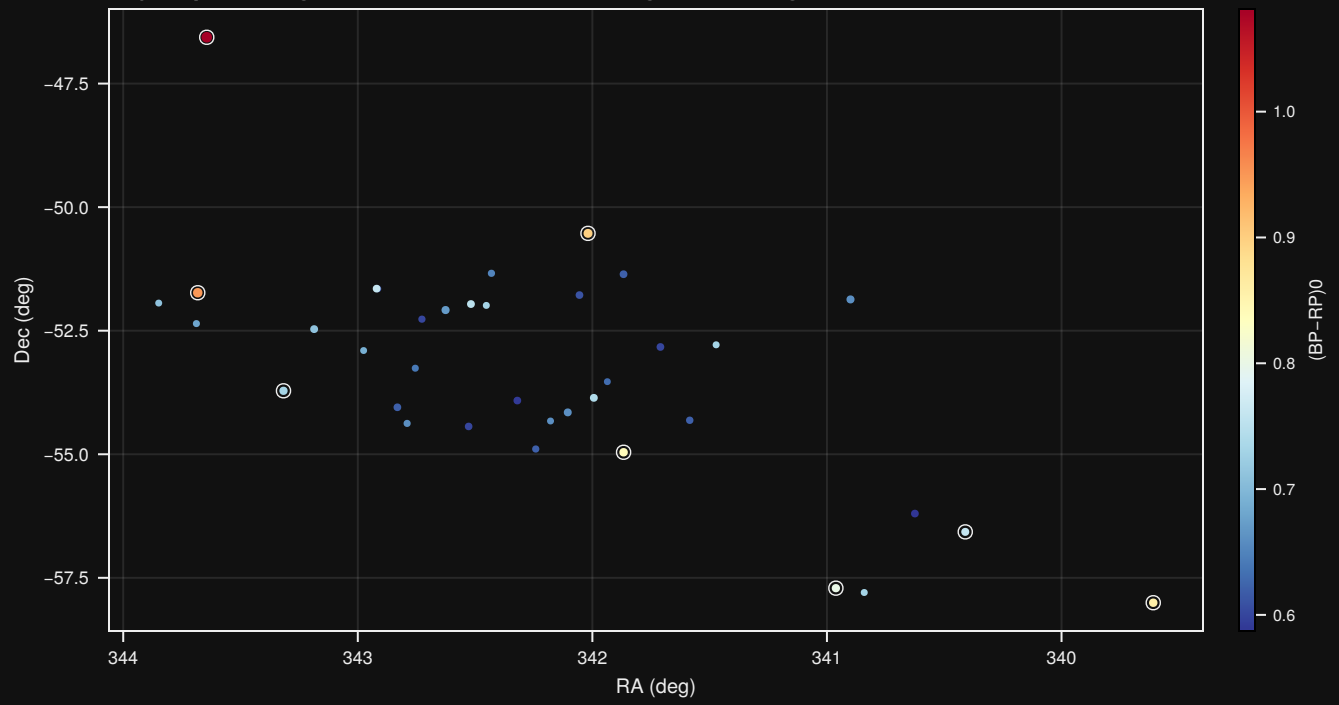
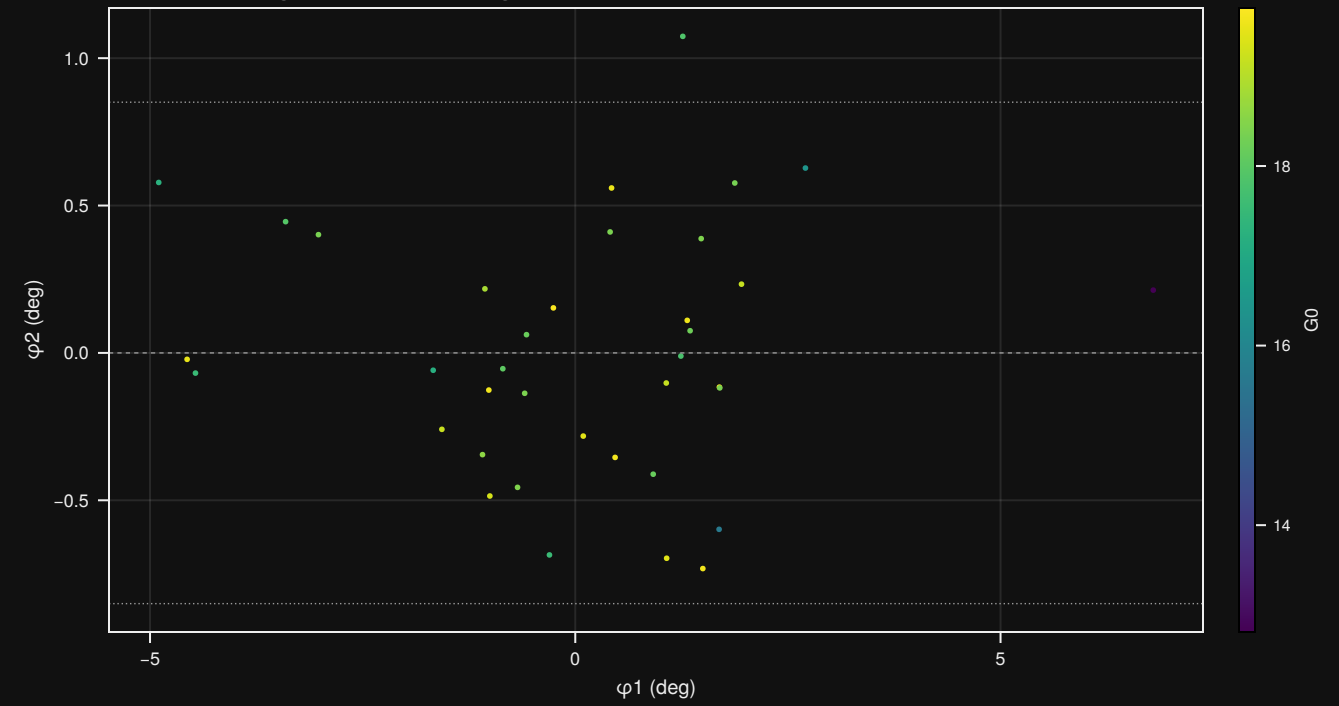
New-27

MEMBERSHIP	
Stars in catalog	37
with Gaia GSP-Phot	22 (59%)
with radial velocity	8
Stream id (sID)	83
GEOMETRY · great-circle fit	
Centroid RA, Dec	342.2, -52.9 deg
Centroid l, b	335.4, -55.6 deg
Pole RA, Dec	260.2, 5.9 deg
Length (φ1, 2–98%)	11.7 deg
Width σ(φ2), robust	0.43 deg
KINEMATICS	
pmRA* med ± σ	1.29 ± 0.38 mas/yr
pmDE med ± σ	-8.56 ± 0.48 mas/yr
μφ1* med ± σ	-8.27 ± 0.49 mas/yr
μφ2 med ± σ	-2.70 ± 0.27 mas/yr
Parallax (cat.) med	0.134 ± 0.231 mas
Parallax (wtd, ZP-corr.)	0.151 ± 0.021 mas
Distance	d = 6.6 ± 0.9 kpc
Physical length	1.36 kpc
RV median ± σ(MAD)	-67.5 ± 130.9 km/s
RV gradient dV/dφ1	-16.09 km/s/deg
PHOTOMETRY · POPULATIONS	
G0 range (median)	12.8–19.8 (18.4)
(BP−RP)0 median	0.67
GSP-Phot [M/H] (n=22)	-1.42 ± 0.53
GSP-Phot Teff median	6026 K
RUWE > 1.4	0.0 %
PM clusters (largest)	2 (14 stars)
IBATA+2024 · TABLE 3	
n catalog | paper	37 | 37
n_RV catalog | paper	8 | 8
<[Fe/H]> paper	-2.08
Stellar mass (paper)	410 Msun
Pole offset vs paper	0.1 deg
Remark	Velocities unclear

Sky map · catalog members (n = 37) · size ∝ brightness · ring = has RV

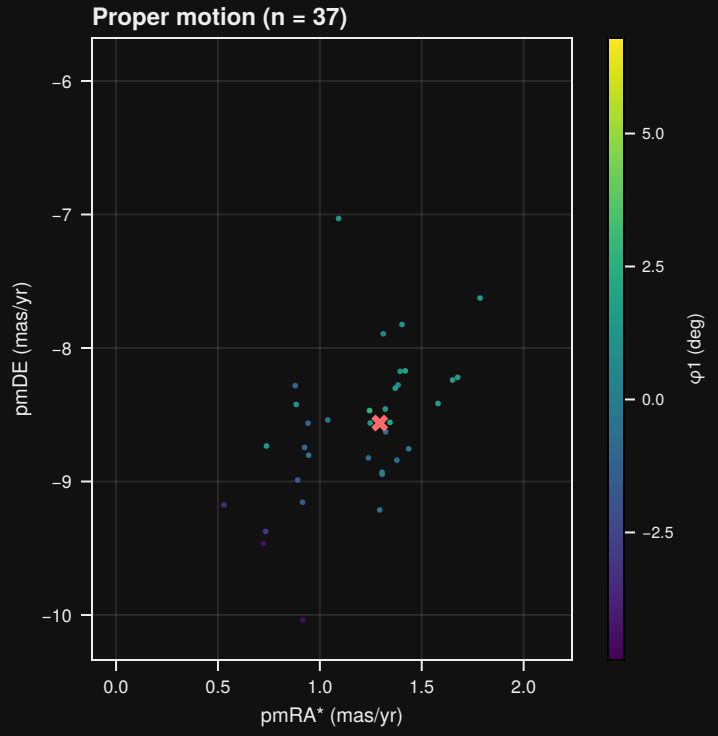


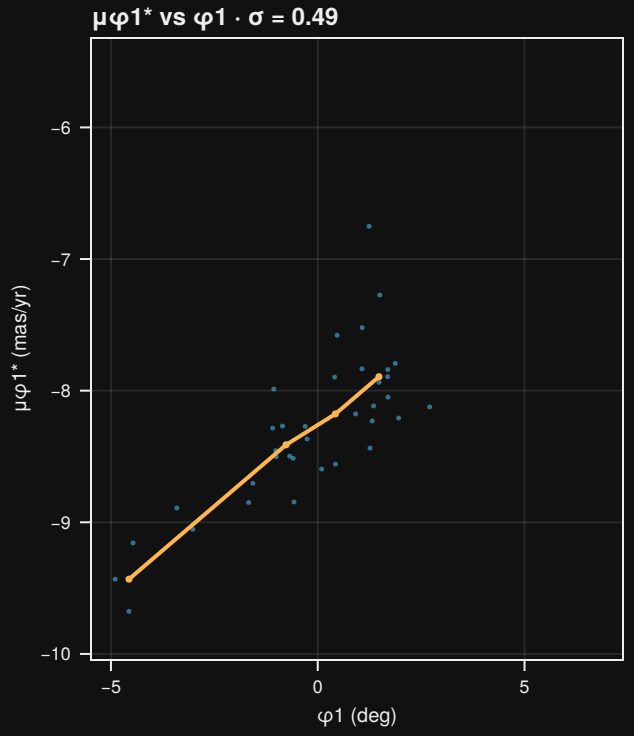
Stream frame (great-circle fit) · length 11.7° · width σ 0.43°

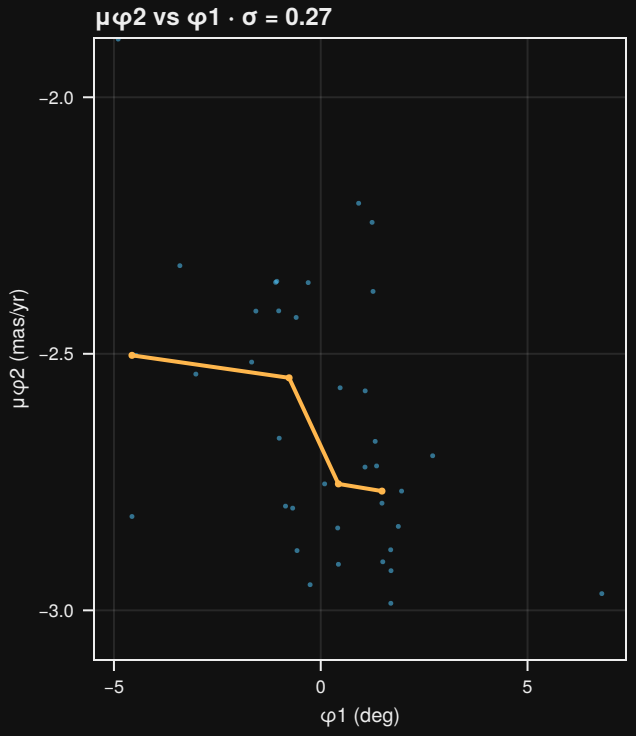


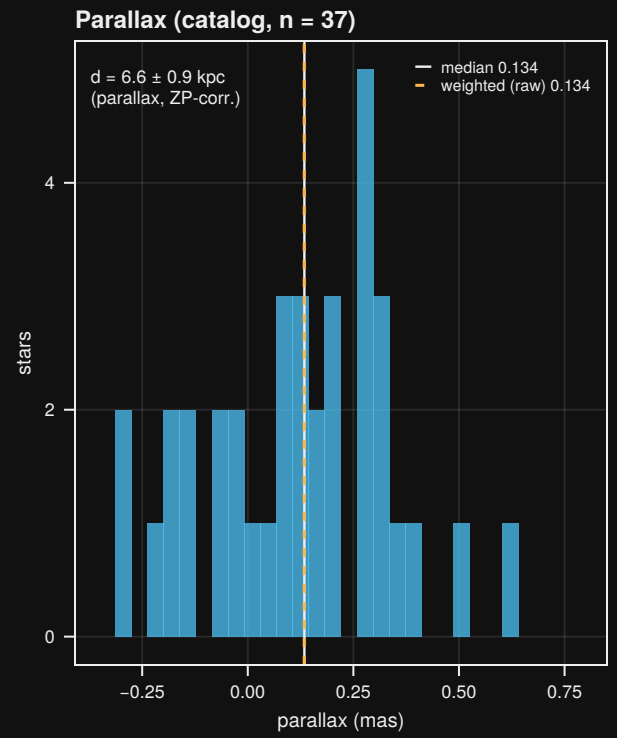
New-27

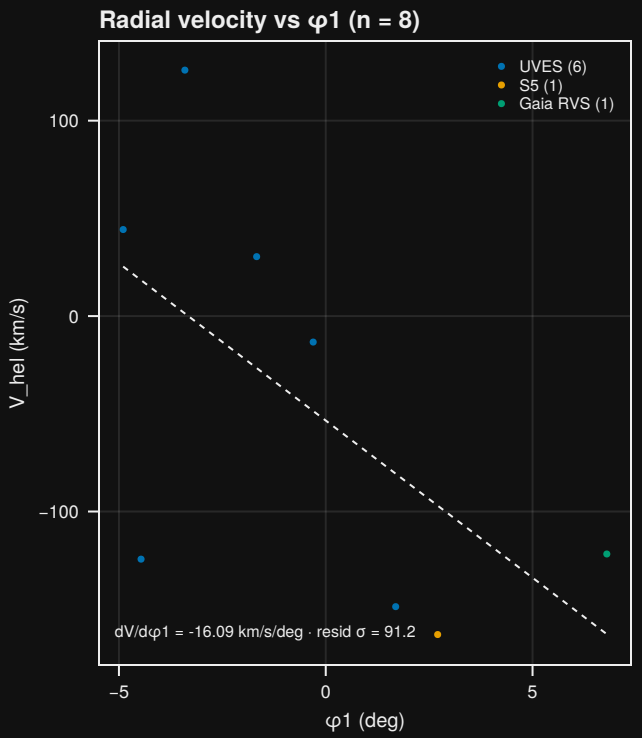
Kinematics · proper motion · parallax · radial velocity

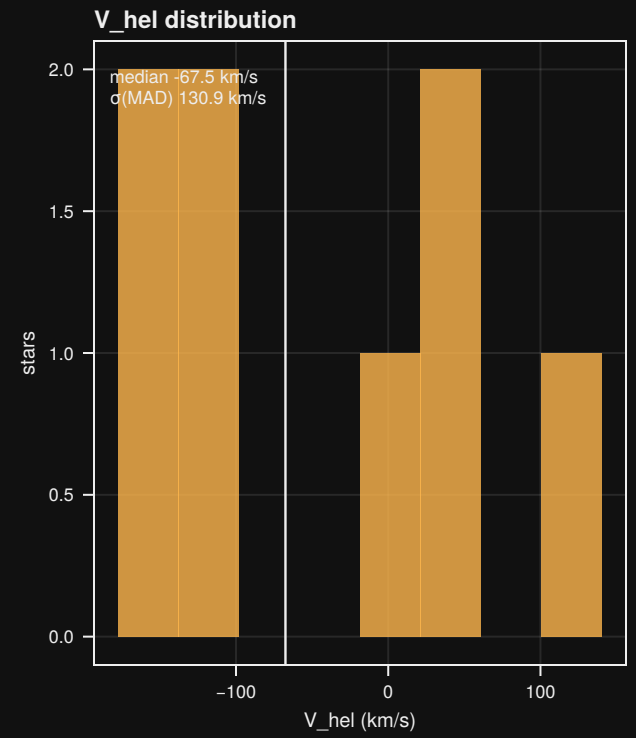






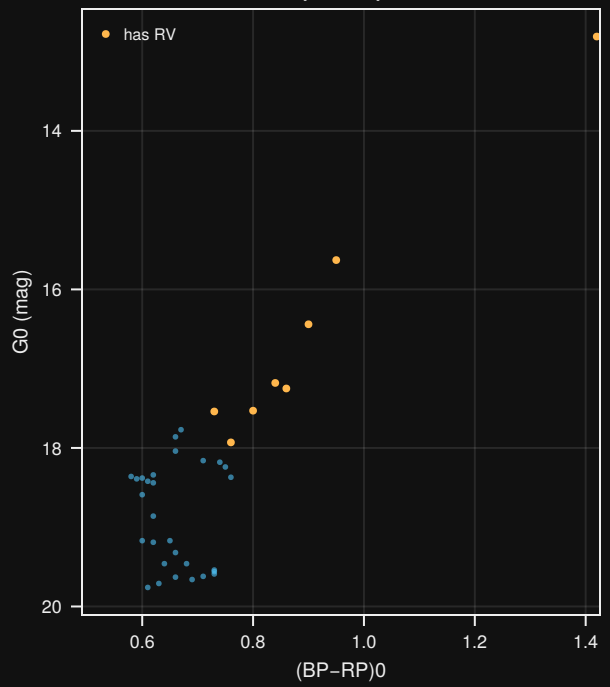




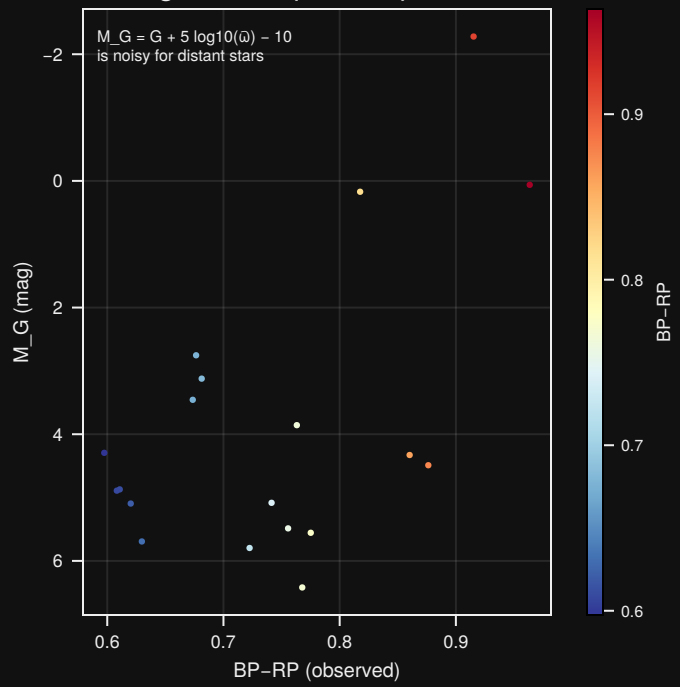


New-27

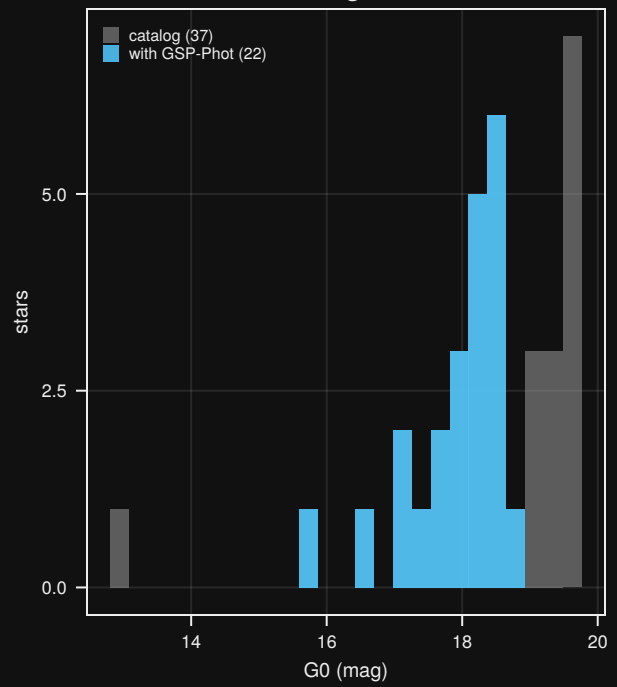
CMD · dereddened (n = 37)



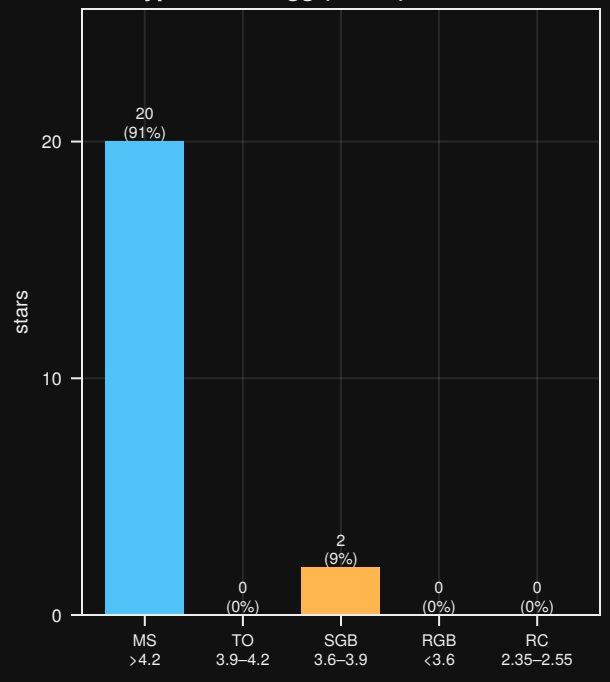
HR diagram · ϖ>0 (n = 19/22)



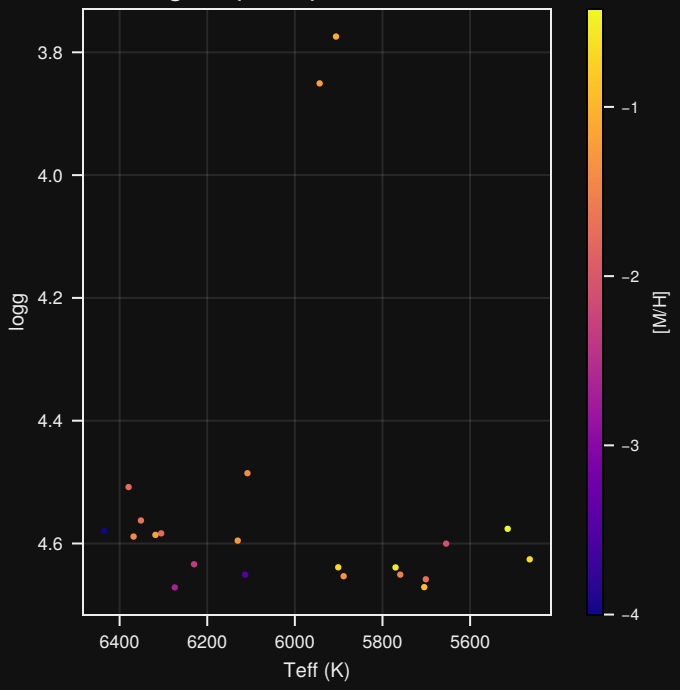
G0 & GSP-Phot coverage



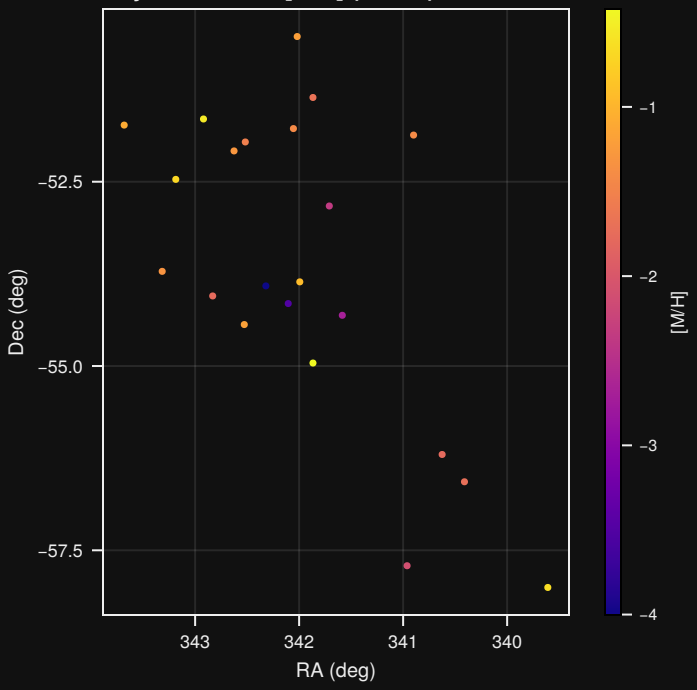
Star types from logg (n = 22)



Kiel diagram (n = 22)



Sky · GSP-Phot [M/H] (n = 22)

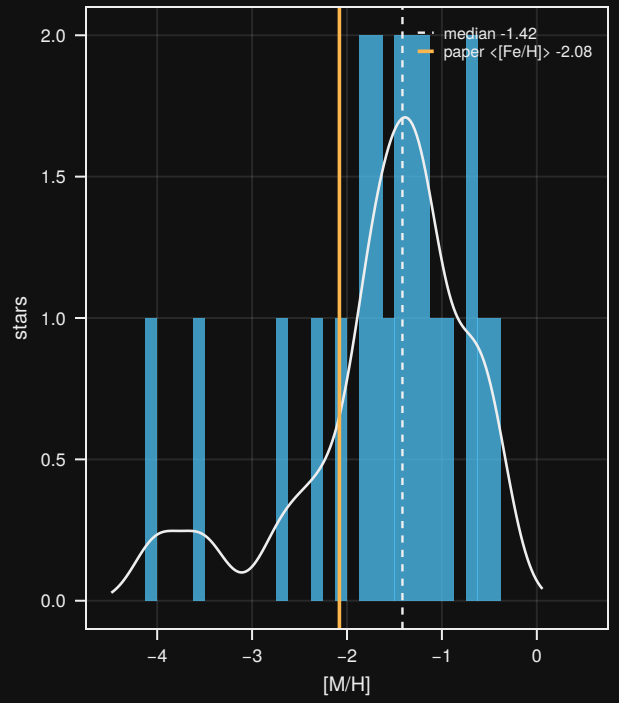


GSP-Phot (Gaia low-resolution BP/RP) parameters are prior-dominated for faint, distant halo stars (G ≥ 17): logg, [M/H] and ages are indicative only and typically biased toward dwarfs.

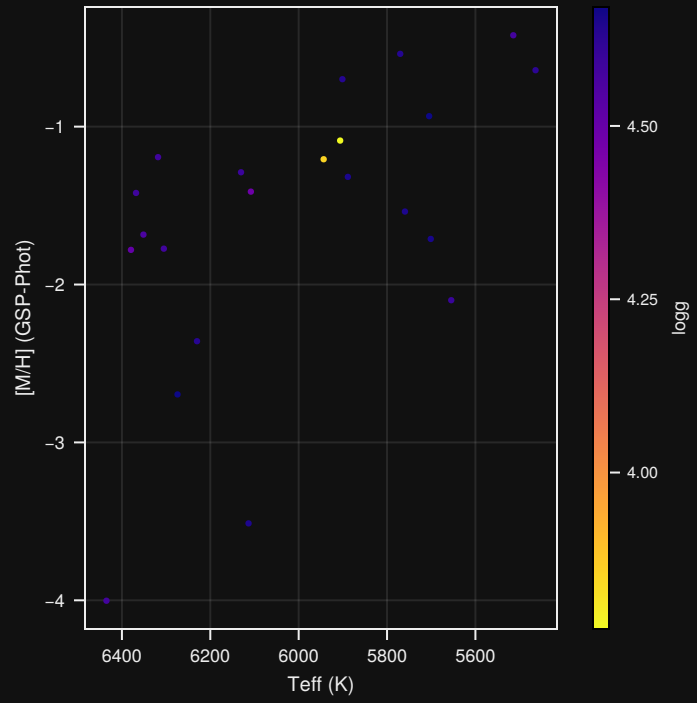
New-27

Metallicity · age · astrometric quality

[M/H] GSP-Phot (n = 22)



Teff vs [M/H] (n = 22)



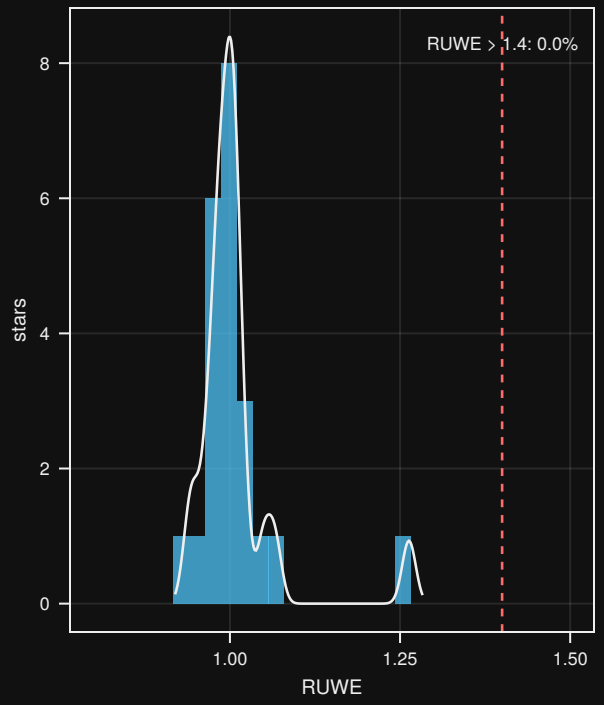
FLAME age (n = 2)

FLAME ages for only 2 star(s)
(FLAME needs good parallaxes)

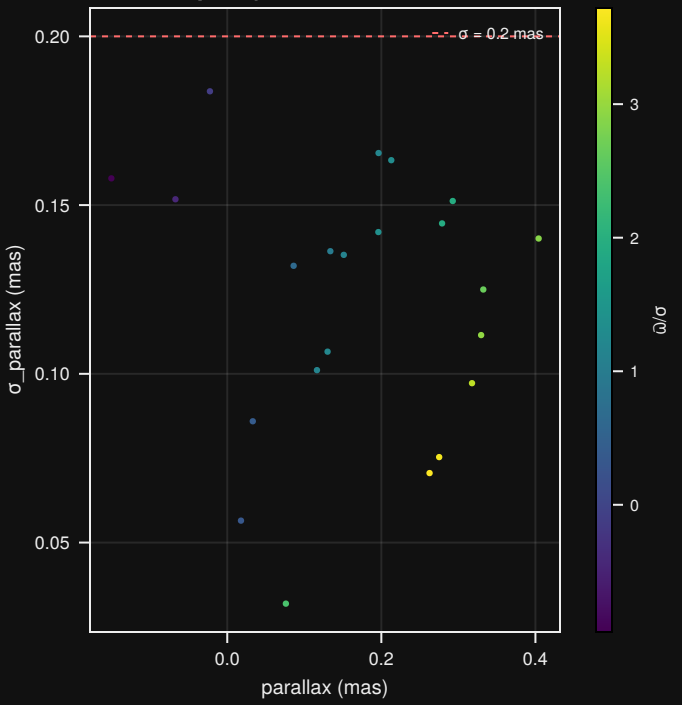
Age–metallicity (n = 2)

n < 3

RUWE (n = 22)

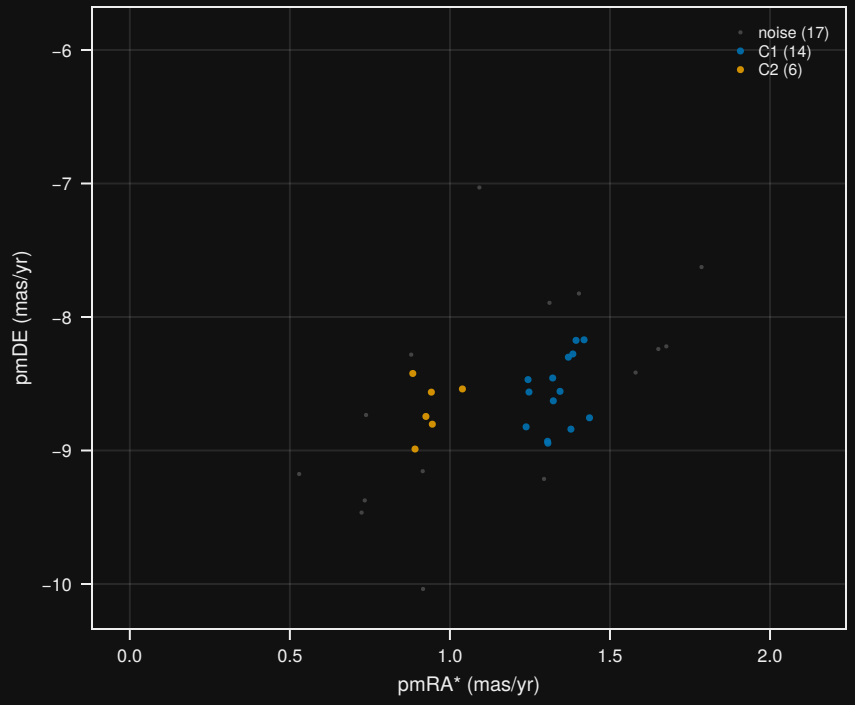


Parallax quality (n = 22)

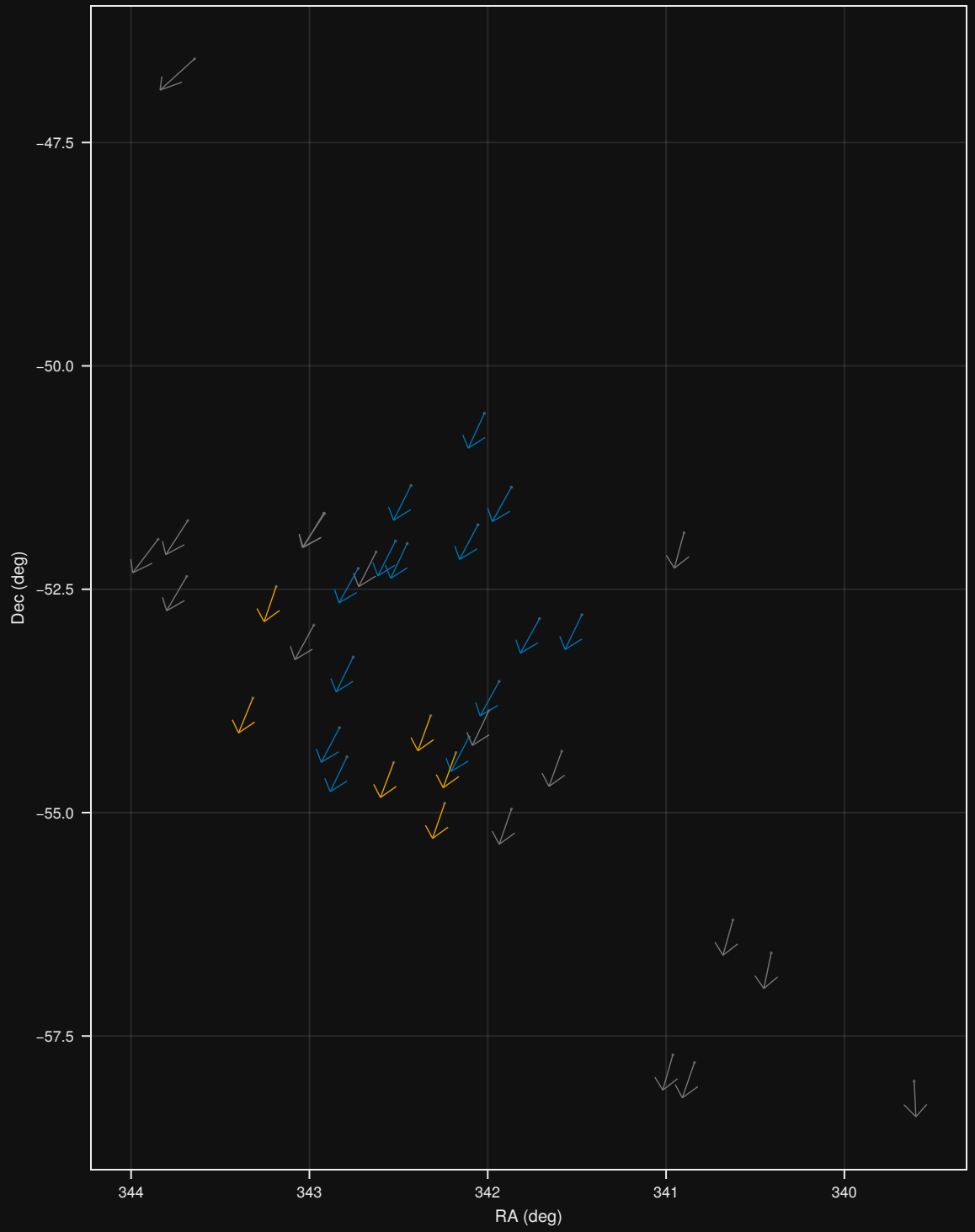


New-27

PM clustering · HDBSCAN · 2 cluster(s) · noise 46%



Proper-motion directions (arrows) coloured by cluster



Cluster summary

id	n	pmRA	pmDE	σpmRA	[M/H]	n_RV
noise	17	1.09	-8.42	0.53	-1.29	6
C1	14	1.33	-8.56	0.07	-1.68	1
C2	6	0.93	-8.65	0.04	-1.26	1

New-27

Per-cluster proper-motion vector maps

